MDB CTS Document — HSA Auger Configuration Tree

Version 1.1 — Revised After Two-Pass Test Review

Evidence Base Differs From Drive Cap and Split Spoon — Read Before Using

This document is built from HSA's extensive audit history, plus a two-pass test review (v0.1, August 2026) that verified two of three proposed corrections against the real Tooling Catalog and Master Repository before adoption. It has still not been traced live against an actual search the way Drive Cap and Split Spoon have. Treat this document as audit-grade plus verified test findings, not fully trace-verified, until a live HSA Auger trace is run.

1. Scope and Entry

Governs: HSA Auger section selection and part number resolution within Workspace 004 (Tooling) — the flagship section of this catalog, including the EXD escalation and Lite-Flite exceptions layered into it.

Trigger: a search or request naming "HSA auger," a specific connection family ("2-Key," "3-Key"), or a size ("3-1/4 auger"), routed per WS004-SEARCH-002. DF001's broader qualification path fires only when the customer has not named a specific HSA product — narrowed scope, confirmed earlier this session.

This document is the executable layer combining the compatibility rules in the Master Repository (HSA-001–004, AUG-001–003, 004.8/004.8.1) with the real part-number data in the Tooling Catalog (Part B, Section 2). Neither source is edited by this document; both are referenced.

2. Governing Rules — Quick Reference

- HSA-001: foundational HSA compatibility — connection family, auger ID, and duty class must all match. Confirmed sound in the contradiction/rule-parity audits.
- HSA-002: a Lead Auger is not a separate tooling system — treat as a compatible auger section plus a compatible cutterhead. See Section 4 below.
- HSA-003: reviewed, confirmed clean.
- HSA-004: if connection type is unknown, ask 2-Key, 2-Key HD, 3-Key, or other before recommending drive cap, cutterhead, or augers.
- AUG-001/002/003: 2-Key SD, 2-Key HD, and 3-Key do not cross-compatible with one another. Confirmed via the non-interchangeability audit — this is a firm boundary, not a soft preference.
- 2-Key SD Connection Variant Rule (confirmed, newly incorporated): 2-Key Standard Duty includes both single-bolt and DB/double-bolt paths. Two bolts does not automatically mean HD — DB is a Standard Duty variant, not a Heavy Duty indicator. A dedicated DB Drive Cap Cross-Reference exists and must be preserved exactly where it applies.
- Lead Assembly Non-Quotable Rule: legacy lead assembly bundled PNs may inform what a first-time setup requires, but BOM/quote output is always individual components — never a bundled lead-flight PN.
- 004.8 / 004.8.1 / CH-001 / CH-003 (Cutterhead Style Neutrality): MDB must never default to Bullet, or any single style, when cutterhead style is unspecified. This was the single largest contradiction found in this entire project. Directly relevant here because most Auger searches lead into a Cutterhead search next.
- STD Duty vs. Hex-Cor: confirmed as separate product types that must never cross-suggest. Relevant to search neutrality if a vague "auger" search could otherwise blend HSA, STD Duty, and Hex-Cor results.

3. Configuration Tree

Four facts specify an HSA Auger section: connection family, connection variant (where applicable), auger ID, and length. Connection family is asked first by default — the same reasoning as Drive Cap's hex-shank-first sequence.

Revised After Test Review — Connection Family Alone Is Not Always Sufficient
"2-Key" alone can leave more than one valid product path where both single-bolt and DB/double-bolt construction exist at the same size. Confirmed directly against the Tooling Catalog's own foundational rules. Connection Variant is now a formal field in this tree, not left to free-text interpretation.

Step	Fact	Values / Behavior	Display Rule
1 (default first)	Connection Family	2-Key SD · 2-Key HD · 3-Key	Order-independent despite default sequence
2 (new)	Connection Variant	Single-Bolt · DB — where the selected family/size supports both	Conditional — hidden where only one construction exists for the resolved family/size
3	Auger ID (Size)	Filtered by family and variant	Retain visibly after selection
4	Length	Standard section lengths per resolved product path	A blank source cell becomes Pending GAP, not a guessed PN

Order independence (per 24.0a) applies exactly as in Drive Cap and Split Spoon: any fact stated first narrows the others without causing them to disappear.

4. Lead Auger Handling (HSA-002)

A request for a "lead auger" does not route to a separate product tree. Per HSA-002, it resolves to two linked results: a compatible auger section (via the tree in Section 3) plus a compatible cutterhead. The cutterhead half of that result

must go through the neutrality handling in Section 6 below — a lead auger request is not license to default to Bullet or any single style.

Per the Lead Assembly Non-Quotable Rule, the output is always individual components, never a bundled legacy lead-assembly PN. Confirmed directly: no cataloged "lead-flight" or "1/2-flight" auger-section variant exists in the real HSA data — the only "1/2 flight" reference in this catalog belongs to a different product (Hex-Cor Series 3000) and does not apply here.

5. Flightless Extension — Separately Routed, Shared Variables

Flightless Extension is not technically an auger section and does not belong in the ordinary Auger terminal matrix. Per WS004-SEARCH-002, a specifically-named product routes to its own configuration path rather than surfacing as a footnote inside a broader one — confirmed as the correct architecture for this item.

This is a routing distinction, not an independence from HSA's core facts. Flightless Extension still requires connection family and auger ID as real, required variables — not because it is an auger, but because it must fit the connection and size already in use on the customer's string. These often come up as one-time purchases driven by job-site constraints, so confirming fit against the existing string matters more here than in a standard multi-item auger order.

- Routing: named directly ("flightless extension," "6-1/4 flightless") — does not require entering the ordinary Auger tree first.
- Required variables: connection family, connection variant where applicable, and auger ID — same facts as Section 3, reused for compatibility, not re-derived from a different source.
- Known data issue, unchanged by this routing decision: PN 150329-34 is recorded for both 2-Key SD and 2-Key HD. AUG-001/002/003 establish these should stay distinct — genuinely unresolved, needs GP verification regardless of which tree it's searched from.

6. Cutterhead Cross-Reference and Neutrality

Most Auger configurations lead directly into a Cutterhead selection — either because the customer asked for a complete lead auger (Section 4) or because they explicitly continue from the auger result. This is the single highest-risk point in the HSA tree, given the confirmed history here.

- When cutterhead style is unspecified, show every style compatible with the resolved connection family and auger ID, grouped by style. Never collapse to Bullet or any other single style by default.
- Compatible styles are filtered by the auger's connection family and ID, established in Section 3 — not re-asked from scratch.
- If the customer states a style directly ("bullet cutterhead"), route to that style specifically — per WS004-SEARCH-002, a named style is specific enough to skip the neutral group-by-style display.

7. Terminal States

State	What the Rep Sees	Next Action
Confirmed PN	The part number, quote-ready pending GP verification	Proceed to quote
Pending GAP	Explicit GAP language from the source table, not a blank result. All constraints stay visible and selected.	Offer a different value to try, or escalate for GP confirmation
Confirmed Non-Offering	A direct statement the combination doesn't exist, with a confirmed reason	No escalation — settled. None currently confirmed for HSA Auger specifically; not to be assumed from Drive Cap's torque findings (case-by-case, per 24.0b).

8. Known Gaps (Real, Specific — Not Assumed)

Flightless Extension's shared-PN conflict is now documented in Section 5, alongside its routing treatment, rather than listed generically here.

- Auger Holders: shared PNs and blank cells in source — pending GAP, needs GP verification. Routing treatment (ordinary matrix vs. separate named-tool tree) not yet decided — open question.
- Spider Bowl: one shared PN across families — pending GAP. The unmachined Blank Bowl entry has been removed from this family's search matrix entirely (production-only item, different workflow). Routing treatment not yet decided — open question.

9. Data Source and Status

- Part numbers: Tooling Catalog, Part B, Section 2 (HSA Backend Component Reference) and Part B Addendum (remaining branches).
- GP verification status: every PN in this document is pending GP verification, consistent with every other part number in this catalog.
- Compatibility rules: Master Repository, HSA-001–004, AUG-001–003, 004.8/004.8.1, CH-001/003, and Section 24 (24.0a, 24.0b).
- Test review basis: MDB HSA Auger CTS Test Findings v0.1 (August 2026) — two of three proposed corrections verified against real source material and adopted (Connection Variant, Flightless Extension routing); one rejected after verification (lead-flight/1/2-flight auger variant — not supported by source, conflicts with the Lead Assembly Non-Quotable Rule).
- Open item, carried forward honestly: this document has not been trace-verified the way Drive Cap and Split Spoon have. A live trace of an actual HSA Auger search — particularly the auger-to-cutterhead handoff in Section 6 — is the recommended next step before treating this document with the same confidence as the first two.